

Literature status: the strictly singular Borel converse

Run `fa_banach_001`; source arXiv:0908.1113

Source question

Beanland's paper *An Ordinal Index on the Space of Strictly Singular Operators* proves that if X and Y are separable Banach spaces and $\mathcal{B}(\mathcal{SS}(X, Y))$ is Borel in $\mathcal{B}(\mathcal{L}(X, Y))$, then there is a countable ordinal ξ such that

$$\mathcal{SS}_\xi(X, Y) = \mathcal{SS}(X, Y).$$

In Section 4.5, “Future Research and Concluding Remarks”, the paper asks whether the converse holds:

Suppose X and Y are separable Banach spaces and there is an $\xi < \omega_1$ such that $\mathcal{SS}_\xi(X, Y) = \mathcal{SS}(X, Y)$. Is $\mathcal{B}(\mathcal{SS}(X, Y))$ a Borel subset of $\mathcal{B}(\mathcal{L}(X, Y))$?

The source states that this problem was communicated by S. Argyros and conjectures a positive answer.

Supporting literature status

Beanland and Dodos' later paper *On strictly singular operators between separable Banach spaces*, arXiv:1006.2672, explicitly restates this conjecture. In its introduction it says that the conjecture appears in [1, §4.5] and that its second main result answers the question negatively.

The relevant theorem is Theorem 4 of [2]:

Theorem 1 (Beanland–Dodos, arXiv:1006.2672, Theorem 4). *For every $1 \leq p < \infty$ there is a Banach space Y_p with an unconditional basis such that:*

1. $\mathcal{SS}(\ell_p, Y_p)$ is a complete coanalytic, in particular non-Borel, subset of $\mathcal{L}(\ell_p, Y_p)$.
2. If $T : \ell_p \rightarrow Y_p$ is strictly singular, then $\varrho(T) \leq 2$.

Why this answers the source question

Definition 1 of [2] is the same Schreier refinement used in the source paper: T is $\mathcal{S}_\xi$ -strictly singular if every normalized basic sequence and every $\varepsilon > 0$ admit a non-empty $F \in \mathcal{S}_\xi$ and a unit vector in the corresponding finite span on which T has norm at most ε . The rank is

$$\varrho(T) = \inf\{\xi : T \text{ is } \mathcal{S}_\xi\text{-strictly singular}\}.$$

Thus $\varrho(T) \leq 2$ for every strictly singular $T : \ell_p \rightarrow Y_p$ implies

$$\mathcal{SS}_2(\ell_p, Y_p) = \mathcal{SS}(\ell_p, Y_p).$$

At the same time, Theorem 4 says that $\mathcal{SS}(\ell_p, Y_p)$ is complete coanalytic and therefore not Borel. This gives separable spaces $X = \ell_p$ and $Y = Y_p$ for which the Schreier index is bounded by 2, but the strictly singular set is non-Borel. The answer to the source question is therefore negative.

Verification notes and limitations

This packet records a known literature answer, not a new proof. The supporting paper explicitly identifies the source conjecture by citing [1, §4.5]. The statement in arXiv:1006.2672 is formulated for $\mathcal{L}(\ell_p, Y_p)$ as a standard Borel space. Since the source paper uses the operator unit ball $\mathcal{B}(\mathcal{L}(X, Y))$, one can restrict the complete coanalytic example to suitable norm balls, or equivalently rescale operators; the non-Borel obstruction is not removed by the unit-ball formulation.

The result does not classify all bounded-index pairs (X, Y) . It only settles the general converse question negatively.

Human review recommendation

Treat arXiv:0908.1113 Section 4.5 as already answered negatively by arXiv:1006.2672. This packet should prevent duplicate work on the same question, but it should not be counted as an original proof from this run.

References

- [1] K. Beanland, *An Ordinal Index on the Space of Strictly Singular Operators*, arXiv:0908.1113.
- [2] K. Beanland and P. Dodos, *On strictly singular operators between separable Banach spaces*, arXiv:1006.2672.